

Ecological imbalances

Ecological imbalance is a condition in which the natural equilibrium of an ecosystem is disturbed, disrupting interactions between living organisms and their environment. Major causes include deforestation, pollution, overpopulation, overexploitation of natural resources, and climate change. Its major consequences include loss of biodiversity, species extinction, disruption of food chains, soil degradation, and increased natural disasters.

Greenhouse Effect and Climate Change

Introduction

The **greenhouse effect** is a natural process in which certain gases in Earth's atmosphere trap heat from the Sun, keeping the planet warm enough to support life. These gases are called **greenhouse gases**, including carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), water vapor, and chlorofluorocarbons (CFCs). Due to increased human activities, the concentration of these gases has risen, intensifying the greenhouse effect and causing **climate change**, which refers to long-term changes in temperature and weather patterns.

Causes

1. Natural Causes

- Volcanic eruptions
- Changes in solar radiation
- Natural release of greenhouse gases

2. Human Causes

- Burning of fossil fuels (coal, petrol, diesel)
 - Deforestation
 - Industrial emissions
 - Transportation (vehicles)
 - Agriculture (methane from livestock, rice fields)
 - Use of CFCs in appliances
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Positive Consequences

- Keeps Earth warm and habitable
 - Supports life processes and agriculture
 - Prevents extreme freezing of the planet
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Negative Consequences

- Global warming
 - Melting of glaciers and ice caps
 - Sea-level rise
 - Extreme weather events (floods, droughts, storms)
 - Loss of biodiversity
 - Health risks and spread of diseases
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Control Measures

- Use renewable energy (solar, wind, hydro)
- Plant more trees (afforestation)
- Reduce fossil fuel use
- Use public transport and eco-friendly vehicles
- Control industrial pollution
- Practice reduce, reuse, recycle
- Avoid CFC-containing products
- Raise environmental awareness

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Acid Rain

Introduction

Acid rain refers to rainfall or any form of precipitation (rain, snow, fog) that contains harmful acids such as sulfuric acid (H_2SO_4) and nitric acid (HNO_3). It is formed when sulfur dioxide (SO_2) and nitrogen oxides (NO_x) released into the atmosphere react with water vapor and oxygen.

Causes

- Burning of fossil fuels in power plants and industries (releases SO_2)
 - Vehicle emissions (release nitrogen oxides)
 - Industrial processes and factories
 - Burning of coal and oil
 - Natural sources like volcanic eruptions (minor cause)
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Consequences

- Damage to crops, forests, and vegetation
 - Acidification of soil (loss of soil fertility)
 - Harm to aquatic life in rivers and lakes
 - Corrosion of buildings, monuments, and metals
 - Health problems such as respiratory issues
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Control Measures

- Reduce use of fossil fuels
- Use clean and renewable energy sources (solar, wind)
- Install scrubbers in factory chimneys
- Use catalytic converters in vehicles
- Enforce strict pollution control laws
- Promote public awareness and eco-friendly practices

Biological Invasions

Introduction

Biological invasion refers to the introduction and spread of non-native (alien) species into a new ecosystem where they establish, spread rapidly, and cause harm to the environment, economy, or human health. These species are called **invasive species**.

Impacts of Biological Invasions

1. Environmental Impacts

- Loss of native biodiversity
- Competition with native species for food and habitat
- Predation on native species
- Disruption of ecosystems and food chains

2. Health Impacts

- Spread of diseases (e.g., invasive mosquitoes spreading malaria, dengue)
- Allergies and skin problems caused by certain plants
- Contamination of water and food sources

3. Social Impacts

- Reduced availability of natural resources
- Impact on livelihoods (farmers, fishers)
- Displacement of local communities dependent on ecosystems

4. Economic Impacts

- Crop damage and reduced agricultural productivity
- Increased cost for control and management
- Damage to fisheries, forestry, and tourism industries

5. Evolutionary Impacts

- Changes in genetic diversity of native species
- Hybridization with native species
- Extinction of species leading to loss of unique genes
- Alteration of natural selection processes

Management of Biological Invasions

1. Prevention

- Strict quarantine laws and border inspections
- Monitoring and early detection of invasive species
- Public awareness and education
- Avoid introduction of exotic species

2. Eradication

- Complete removal of invasive species at early stages
- Destruction of breeding populations
- Use of targeted methods before species spread widely

3. Control Methods

a. Manual Control

- Hand picking or removal of invasive plants and animals
- Hunting or trapping

b. Mechanical Control

- Use of machines (cutting, mowing, plowing)
- Physical barriers and traps

c. Chemical Control

- Use of pesticides, herbicides, and insecticides
- Must be used carefully to avoid environmental harm

d. Biological Control

- Introduction of natural predators, parasites, or pathogens
- Helps reduce invasive species population naturally

Vegetation – Introduction

Vegetation refers to the plant life growing in a particular area, including trees, shrubs, grasses, and other types of plants. It is a natural cover of plants that develops in a region based on factors like climate, soil, and topography. Vegetation plays an important role in maintaining ecological balance, providing habitat for animals, and supporting human life.

Types of Vegetation in Nepal

Nepal's vegetation varies with altitude, climate, and rainfall, ranging from tropical lowlands to alpine highlands.

1. Tropical Region (Below 1,000 m)

Sal Forest (*Shorea robusta*) – Terai region, dense sal trees with shrubs and grasses.

Tropical Deciduous Riverine Forest – Along rivers; species include sissoo, khair.

Tropical Evergreen Forest – Found in high rainfall areas; species include mango, jamun, mahogany.

2. Subtropical Region (1,000–2,000 m)

Chilaune-Katus Forest – Chilaune (*Schima wallichii*) and katus (*Castanopsis indica*).

Chirpine Forest – Dominated by chirpine (*Pinus roxburghii*) on dry slopes.

Utis Forest – Dominated by utis (*Alnus nepalensis*), common on moist hillsides.

3. Temperate Region (2,000–3,000 m)

Lower Temperate Mixed Broad-Leaved Forest – Oak, maple, rhododendron, and other broad-leaved trees.

Temperate Mixed Evergreen Forest – Evergreen species mixed with broad-leaved trees; includes pine, fir, spruce.

Broad-Leaved Oak Forest – Dominated by various oak species; found on cooler, moist slopes.

Coniferous Forest – Dominated by pine, fir, and spruce; adapted to colder, higher altitudes.

Upper Temperate Mixed Broad-Leaved Forest – Mixture of broad-leaved and coniferous trees; found near upper temperate limits.

4. Sub-Alpine Region (3,000–4,000 m)

Silver Fir Forest – Dominated by *Abies spectabilis* (silver fir).

Birch-Rhododendron Forest – Stunted birch trees with rhododendron shrubs.

5. Alpine Region (Above 4,000 m)

Juniper-Rhododendron Forest – Low shrubs of juniper and rhododendron.

Caragana-Lonicera Forest – Shrubland dominated by *Caragana* and *Lonicera* species.

Alpine Meadow – Grasses, mosses, lichens, and medicinal herbs adapted to cold, harsh climate.

Biodiversity conservation

Biodiversity refers to the variety of life on Earth, including species of plants, animals, microorganisms, and the ecosystems they form. **Biodiversity conservation** is the practice of protecting, preserving, and managing this diversity to maintain ecological balance, ensure the survival of species, and sustain human life.

Conservation is essential because biodiversity provides **food, medicine, clean air and water, raw materials**, and helps regulate climate and ecosystems. Loss of biodiversity due to habitat destruction, pollution, overexploitation, invasive species, and climate change can threaten the health of the planet and human well-being.

Biodiversity conservation can be achieved through **in-situ methods** (protecting species in their natural habitats, like national parks) and **ex-situ methods** (protecting species outside their natural habitats, like botanical gardens and zoos).

Biodiversity Conservation Methods

1. In-Situ Conservation

Definition: Protecting species in their natural habitats. Examples include **national parks, wildlife reserves, and biosphere reserves**.

Advantages:

- Maintains ecological balance and natural interactions
- Conserves a large number of species at once
- Protects entire ecosystems, not just individual species
- Allows natural evolution and adaptation

Disadvantages:

- Difficult to protect species from poaching, habitat destruction, or natural disasters
- Requires large areas of land, which may be expensive or difficult to manage
- May not help species that are critically endangered and have very few individuals

2. Ex-Situ Conservation

Definition: Protecting species outside their natural habitats. Examples include **zoos, botanical gardens, seed banks, and tissue culture labs**.

Advantages:

- Useful for critically endangered species
- Easier to monitor and manage individual species
- Can be used for research and breeding programs to reintroduce species into the wild

Disadvantages:

- Expensive to maintain and manage
- Cannot replicate natural habitat completely, affecting natural behaviors
- Limited number of species can be conserved at a time

Natural Environment, Vegetation, and Human Activities

Introduction

The **natural environment** includes land, water, air, climate, and living organisms like plants and animals. **Vegetation** is the plant life that grows naturally in an area and plays a vital role in maintaining ecological balance, preventing soil erosion, supporting wildlife, and regulating climate.

Human activities, however, can significantly affect vegetation, leading to environmental degradation and ecological imbalance.

Man-Made Causes of Destruction of Vegetation

1. **Deforestation** – Clearing forests for agriculture, settlements, and industries.

2. **Urbanization and Infrastructure Development** – Roads, cities, and dams destroying natural habitats.
3. **Overgrazing** – Livestock feeding excessively, damaging natural vegetation.
4. **Mining and Quarrying** – Removal of vegetation for minerals and construction materials.
5. **Pollution** – Air, water, and soil pollution affecting plant growth.
6. **Unsustainable Agriculture** – Monoculture, excessive use of chemical fertilizers, and land clearing.
7. **Introduction of Invasive Species** – Non-native plants outcompeting local vegetation.

Hazards / Effects of Destruction of Vegetation

Environmental Hazards

- Soil erosion and desertification
- Increased flooding due to loss of natural water absorption
- Reduced rainfall and microclimate changes
- Loss of biodiversity and wildlife habitats

Economic and Social Effects

- Reduced agricultural productivity
- Loss of timber, medicinal plants, and other forest products
- Increased vulnerability to natural disasters (landslides, droughts)
- Impact on livelihoods dependent on forests and vegetation

Health and Climatic Effects

- Poor air quality due to fewer trees filtering pollutants
- Contribution to global warming and climate change
- Increased heat and reduced shade in urban areas

Virus

Introduction

A **virus** is a microscopic infectious agent that can replicate only inside the living cells of a host. Viruses infect animals, plants, bacteria, and other microorganisms, causing a wide range of diseases in humans, animals, and plants.

Characteristics of Viruses

1. Extremely small (20–300 nm), visible only under an electron microscope.
2. Contain either **DNA or RNA**, but never both.
3. Surrounded by a **protein coat** called a capsid; some have an outer lipid envelope.
4. Reproduce only inside living host cells (**obligate intracellular parasites**).
5. Lack cellular structures such as cytoplasm, nucleus, or organelles.
6. Infect specific hosts due to surface proteins.

Nature of Virus: Living or Non-Living?

Living Characters:

- Have genetic material (DNA or RNA)
- Can reproduce inside host cells
- Can evolve over time

Non-Living Characters:

- Do not have a cellular structure
- Cannot metabolize or produce energy on their own
- Cannot reproduce outside host cells

Conclusion: Viruses are considered “**at the border of living and non-living**”.

Structure of a Virus

1. **Nucleic Acid:** DNA or RNA (single-stranded or double-stranded).
 2. **Capsid:** Protein coat surrounding the nucleic acid, made of capsomeres.
 3. **Envelope:** Some viruses have an outer lipid envelope with glycoprotein spikes.
 4. **Special Structures:**
 - **Bacteriophage:** Head (nucleic acid), tail, and tail fibers to attach to bacteria.
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Economic Importance of Viruses**Advantages:**

- Used in **genetic engineering and biotechnology** (e.g., gene therapy, vaccines).
- Can serve as **biological control agents** to manage insect pests and weeds.
- Important in **research** for understanding molecular biology and cell functions.

Disadvantages:

- Cause diseases in humans, animals, and plants.
 - Lead to **economic losses** in agriculture and livestock.
 - Hard to control and treat once infection occurs.
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Viral Diseases in Plants

Disease	Causative Virus	Symptoms
Tobacco Mosaic	Tobacco Mosaic Virus (TMV)	Mosaic patterns on leaves, stunted growth
Cauliflower Mosaic	Cauliflower Mosaic Virus (CaMV)	Stunted growth, leaf deformation
Banana Bunchy Top	Banana Bunchy Top Virus (BBTV)	Bunched and distorted leaves
Potato Virus Y	Potato Virus Y (PVY)	Leaf mottling, reduced yield

Disease	Causative Virus	Symptoms
Rice Tungro	Rice Tungro Bacilliform Virus + Spherical Virus	Yellowing, stunted plants

Viral Diseases in Animals and Humans

Disease	Causative Virus	Host / Symptoms
Influenza	Influenza Virus	Fever, cough, body pain
HIV/AIDS	Human Immunodeficiency Virus (HIV)	Immune system failure
Chickenpox	Varicella-Zoster Virus	Rash, fever, itching
Measles	Measles Virus	Fever, rash, cough
Rabies	Rabies Virus	Neurological symptoms, death
Foot-and-Mouth Disease	Aphthovirus	Blisters in cattle, pigs, sheep
Rinderpest	Rinderpest Virus	Fever, diarrhea in cattle
Feline Leukemia	Feline Leukemia Virus	Immunosuppression in cats
Smallpox (eradicated)	Variola Virus	Fever, pustules, scarring
Dengue	Dengue Virus	Fever, joint pain, hemorrhage
Hepatitis B	Hepatitis B Virus (HBV)	Liver infection, jaundice